

DATA WITH BARAA

SQL DATA Analytics

Project

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DATA WITH BARAA
SQL Course | Data Analytics Project



SQL Projects

Data Warehousing

"Organize, Structure, Prepare"

- ETL/ELT Processing
- Data Architecture
- Data Integration
- Data Cleansing
- Data Load
- Data Modeling

Exploratory Data Analysis (EDA)

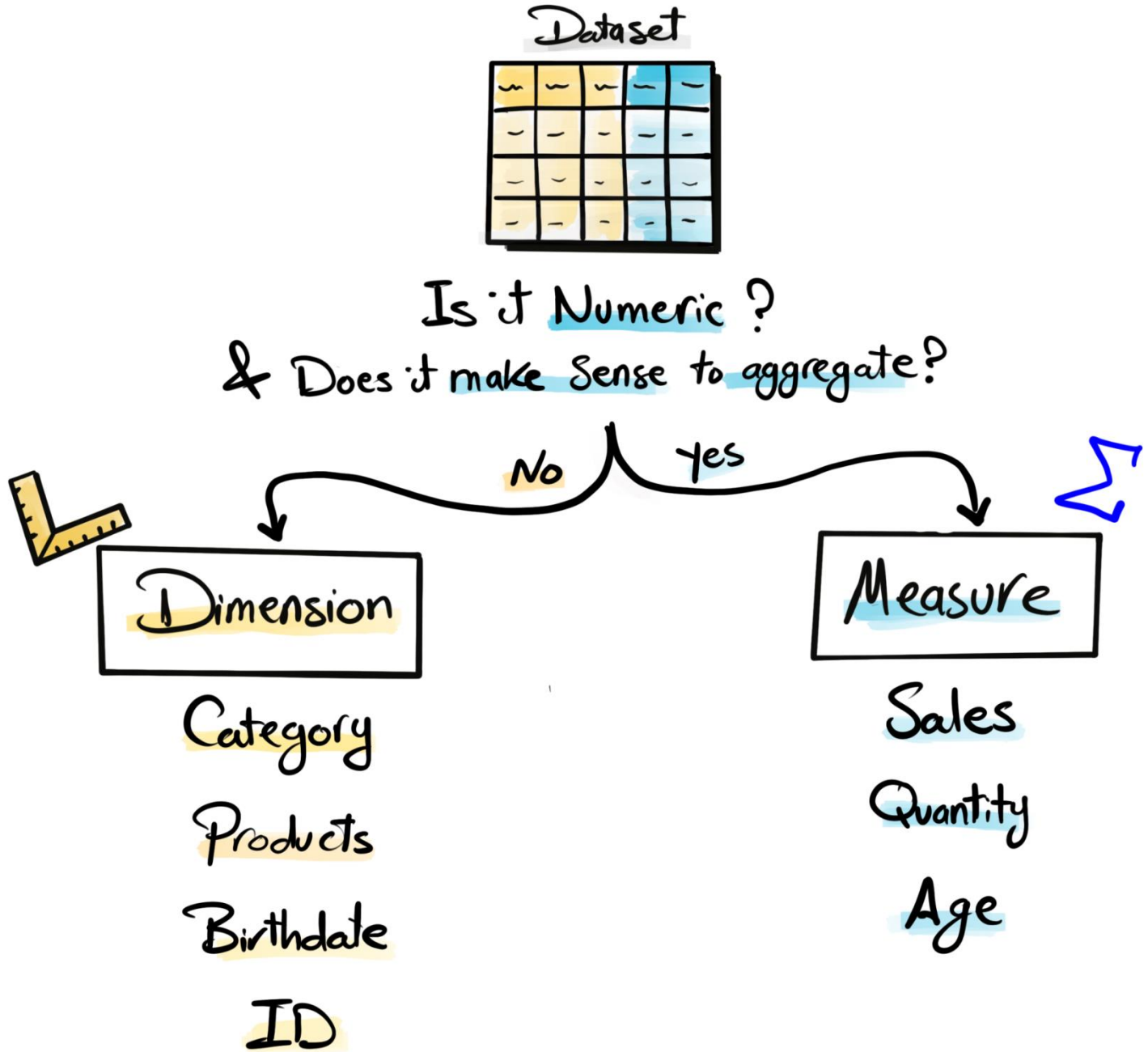
"Understand Data"

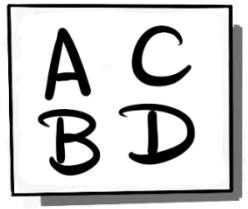
- Basic Queries
- Data Profiling
- Simple Aggregations
- Subquery

Advanced Data Analytics

"Answer Business Questions"

- Complex Queries
- Window Functions
- CTE
- Subqueries
- Reports

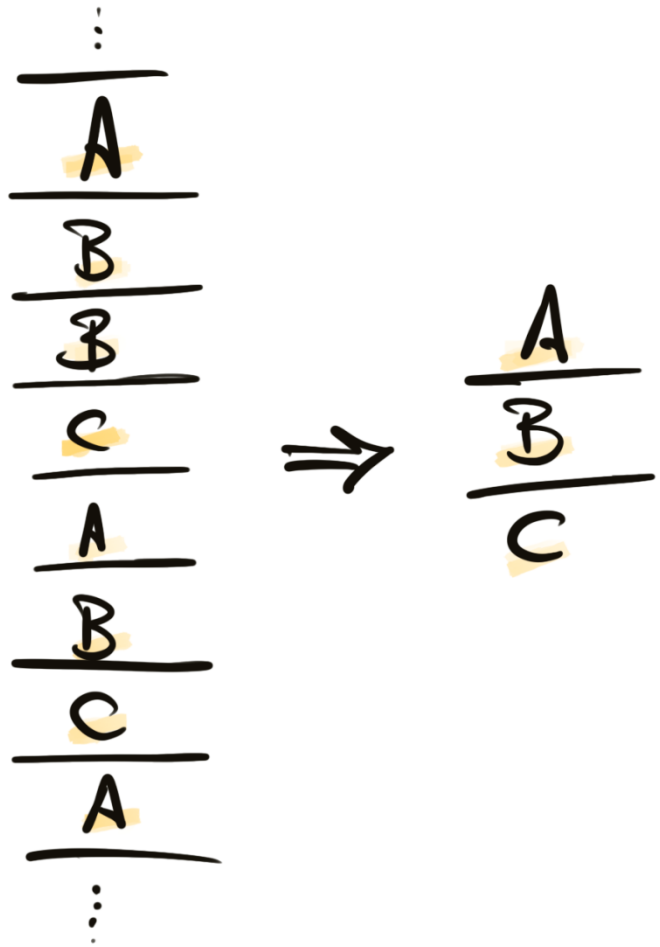

Dataset
Is it Numeric ?
& Does it make Sense to aggregate?
No
Yes
Dimension
Category
Products
Birthdate
ID
Measure
Sales
Quantity
Age

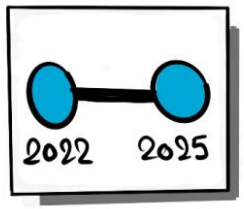


Dimensions Exploration

DISTINCT [Dimension]

DISTINCT Country
DISTINCT Category
DISTINCT Product



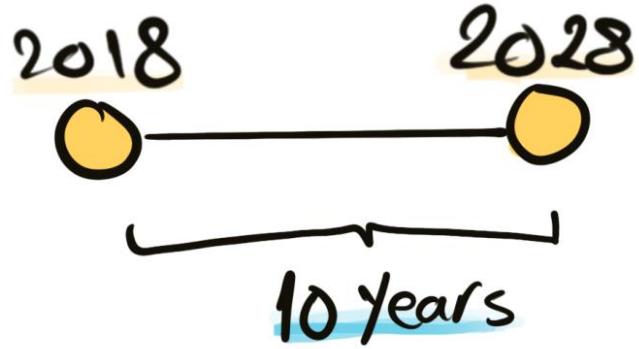


Date Exploration

MIN/MAX [Date Dimension]

- MIN Order_date
- MAX Create_date
- MIN Birthdate

2019
2020
2018
2018
2022
2023
2023
2028
2022



DATEDIFF

999

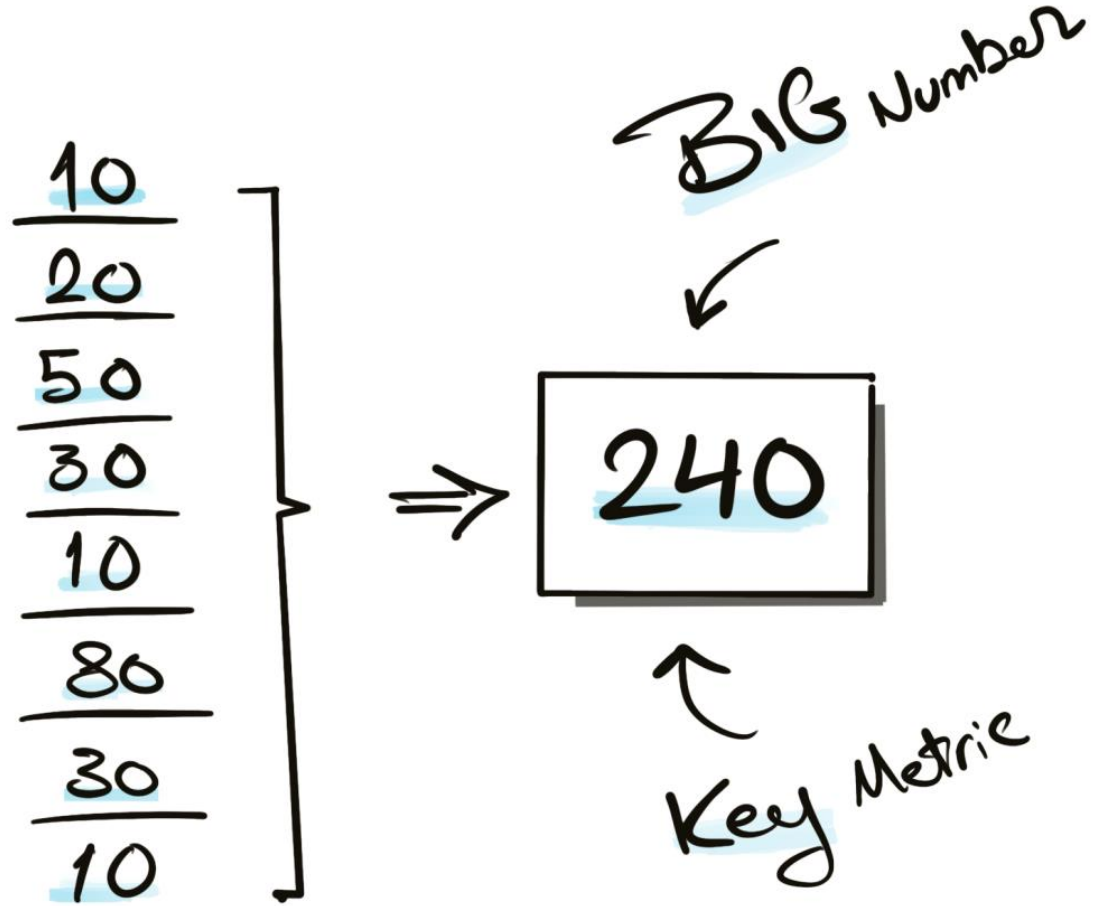
Measures Exploration

$\sum$ [Measure]

SUM (Sales)

AVG (Price)

SUM (Quantity)

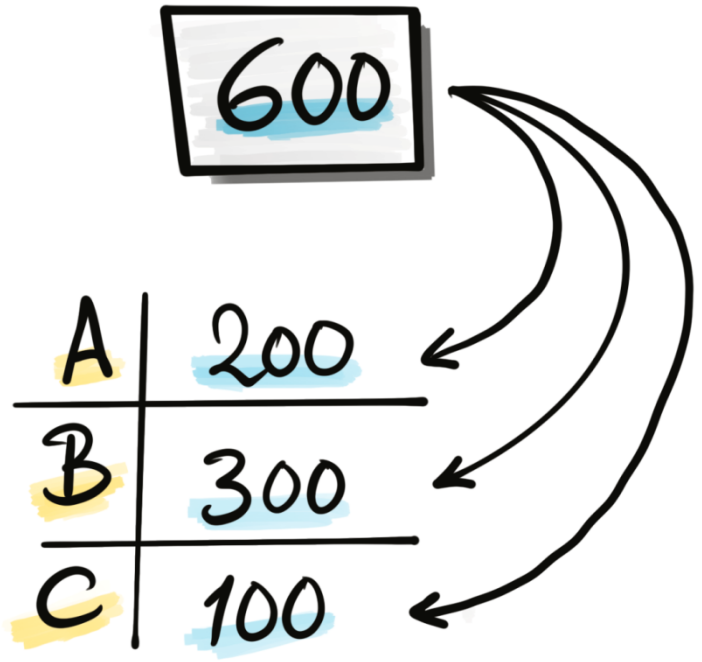


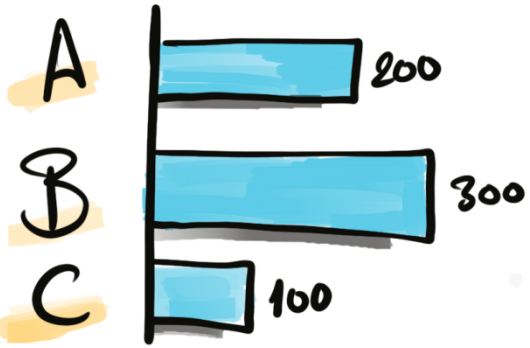


Magnitude

$\sum$ [Measure] By [Dimension]

- Total Sales By Country
- Total Quantity By Category
- Average Price By Product
- Total Orders By Customer







Change-Over-Time Trends

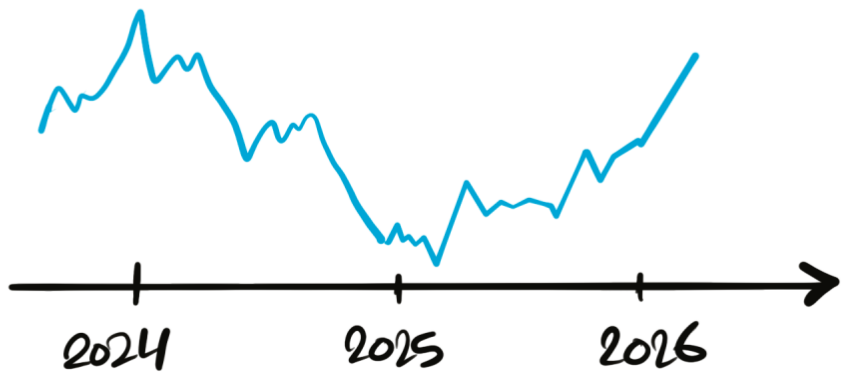
DATA WITH BARAA

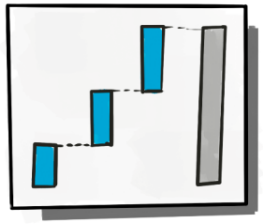
$\sum$ [Measure] By [Date Dimension]

Total Sales By Year

Average Cost By Month

2024	300
2025	100
2026	200



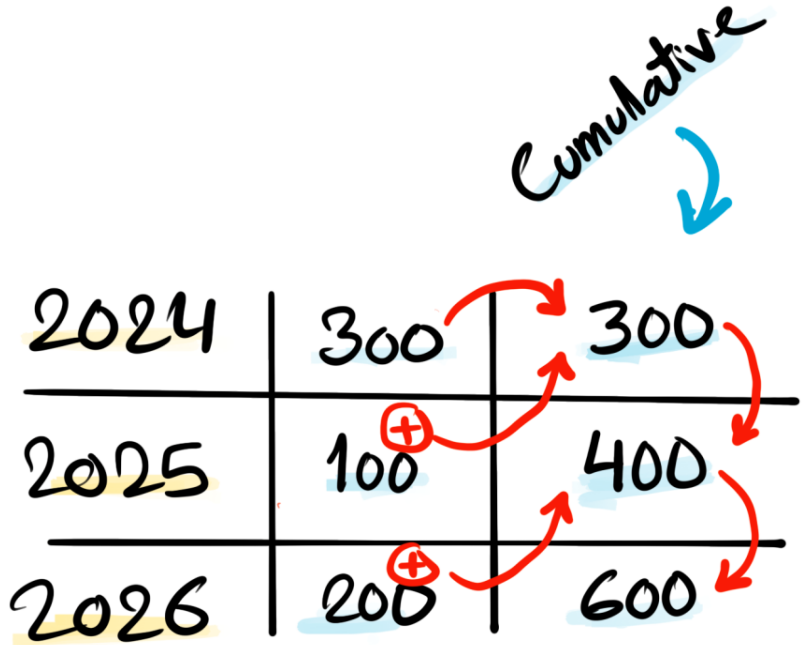


Cumulative Analysis

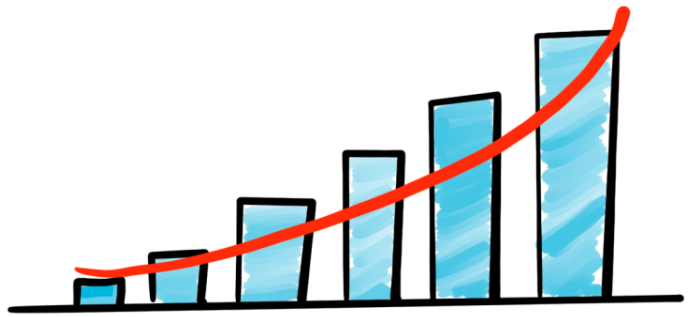
∑ [Cumulative Measure] By [Date Dimension]

Running Total Sales By Year

Moving Average of Sales By Month



WINDOW FUNCTIONS





Performance Analysis

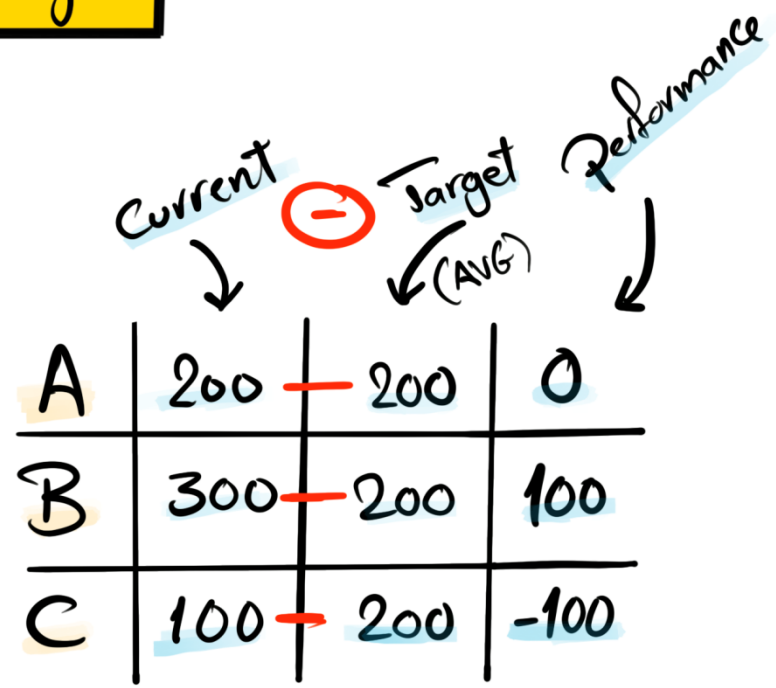
DATA WITH BARAA

Current [Measure] – Target [Measure]

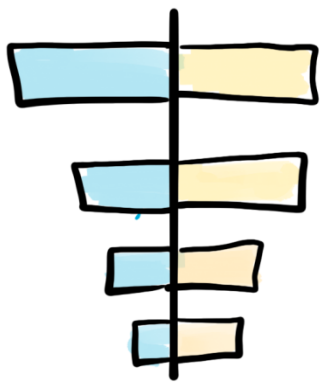
Current Sales – Average Sales

Current Year Sales – Previous Year Sales

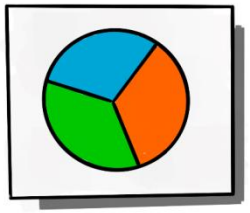
Current Sales – lowest Sales



A	200	200	0
B	300	200	100
C	100	200	-100



WINDOW FUNCTIONS



Part-to-Whole

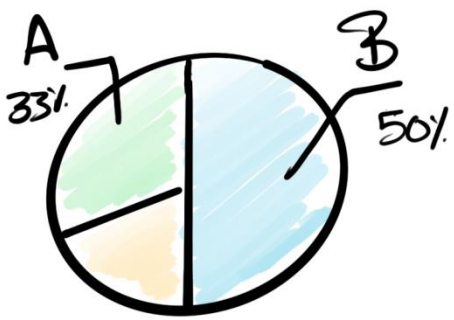
Proportional Analysis

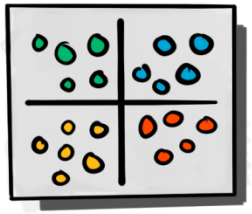
([Measure] / Total [Measure]) * 100 By [Dimension]

(Sales / Total Sales) * 100 By Category

(Quantity / Total Quantity) * 100 By Country

A	200	33%
B	300	50%
C	100	17%





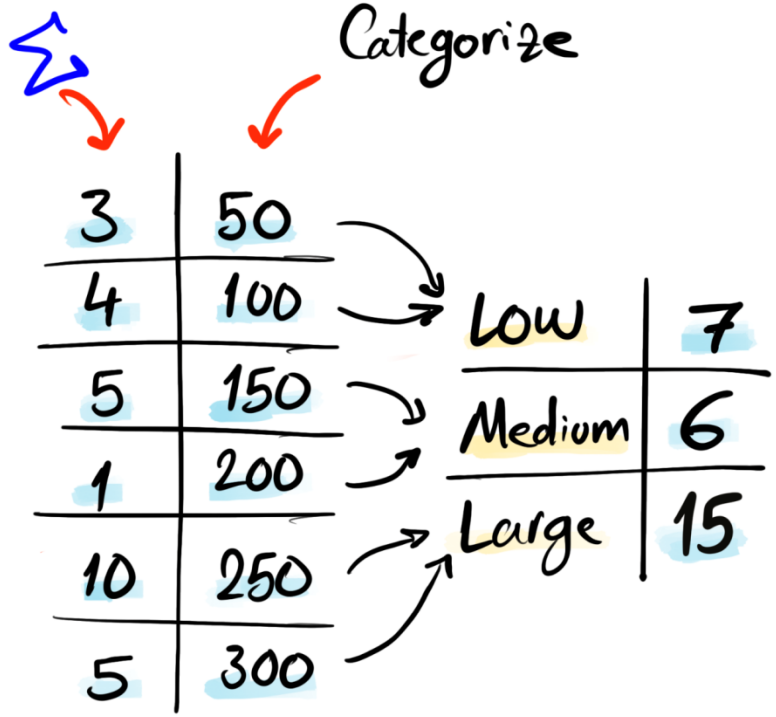
Data Segmentation

DATA WITH BARAA

[Measure] By [Measure]

Total Products By Sales Range

Total Customers By Age



CASE WHEN STATEMENT

